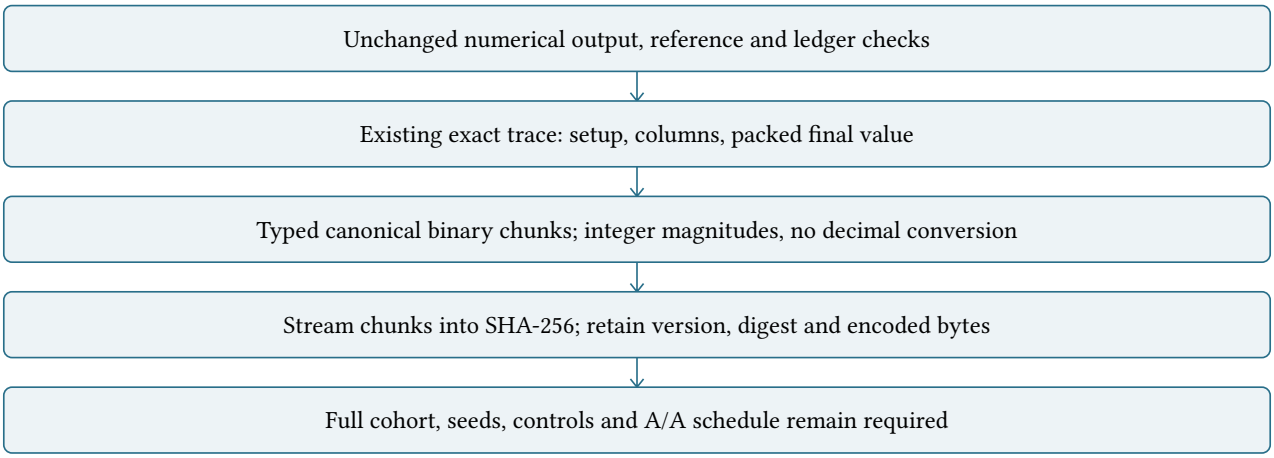
Exact streaming trace fingerprints

2026-11-01 · Tested software component; no kernel trial or timing benchmark

Inspected path. The frozen checker serializes each complete packed-integer trace to sorted JSON solely for a SHA-256 fingerprint. The string is then discarded. This is confirmed from source; its contribution to the original cap has not been measured. The original invocation remains terminal and unchanged: candidate complete, A/A partial, native exits 124.



Implemented alternative. trace_codec.py emits a domain/version header, distinct type tags, canonical lengths and binary integer magnitudes. Strict UTF-8 strings, lists and sorted string-key dictionaries are supported. None, booleans and integers remain distinct. A separately implemented decoder rejects malformed or noncanonical encodings, truncation and trailing bytes. The fingerprint includes schema, SHA-256 and encoded byte count. The full trace object remains in memory; no whole JSON trace string is constructed.

Correctness argument. Type tags and self-delimiting canonical payloads identify each value and container boundary uniquely. Sorted distinct keys fix dictionary order. This establishes encoding injectivity over supported values, not cryptographic collision freedom. SHA-256 retains its usual fingerprint assumption. The new fingerprint is not the old JSON hash.

Software checks completed. Eight deterministic unittest cases pass, covering 100,001-bit integers, signs/zero, Unicode, empty/nested containers, handwritten golden bytes, canonical ordering, type/boundary distinctions, every truncation of a structured fixture, malformed encodings and depth guards. Streamed hashes match hashes of materialized encodings. No numerical kernel was imported. The reported test duration is not used as performance evidence.

Integration boundary. integration.patch is unapplied. Whole proposed source passes syntax checks. AST checks establish unchanged main schedule, stage enumeration and verify statements before its final returned record. It retains the declared 1,698 rows, 34,248 products, both arms, all inputs, references and controls. Only imports and the trace-fingerprint representation change. Original/proposed source hashes are in integration.json.

Next. Add version/hash/byte-field validation to a fresh analyzer and freeze a complete separate instrument contract, locks, seed table and finite budget before any execution. Preserve original capped records and every missing A/A case. No kernel arithmetic or measured speed improvement is claimed by this codec change. DESIGN.md records the exact encoding and rejection rules.